2016.0 RANGE ROVER (LG), 211-04

STEERING COLUMN

DIAGNOSIS AND TESTING

PRINCIPLES OF OPERATION

For a detailed description of the steering wheel heater, refer to the relevant Description and Operation section in the workshop manual.
REFER to: Steering Column (211-04 Steering Column, Description and Operation).

INSPECTION AND VERIFICATION

CAUTIONS:

- Diagnosis by substitution from a donor vehicle is **NOT** acceptable. Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle.
- When probing connectors to take measurements in the course of the pinpoint tests, use the adapter kit, part number 3548-1358-00.

NOTES:

- If the control module or a component is suspect and the vehicle remains under manufacturer warranty, refer to the Warranty Policy and Procedures manual (section B1.2), or determine if any prior approval programme is in operation, prior to the installation of a new module/component.
- Generic scan tools may not read the codes listed, or may read only 5-digit codes. Match the 5 digits from the scan tool to the first 5 digits of the 7-digit code listed to identify the fault (the last 2 digits give extra information read by the manufacturer-approved diagnostic system).
- When performing voltage or resistance tests, always use a digital multimeter accurate to three decimal places, and with an up-to-date calibration certificate. When testing resistance always take the resistance of the digital multimeter leads into account.
- Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests.
- Inspect connectors for signs of water ingress, and pins for damage and/or corrosion.
- If DTCs are recorded and, after performing the pinpoint tests, a fault is not present, an intermittent concern may be the cause. Always check for loose connections and corroded terminals.

1. Verify the customer concern

2. Visually inspect for obvious signs of damage and system integrity

Visual Inspection

ELECTRICAL
▪ Fuses ▪ Wiring harness

- Loose or corroded connector(s)
- Steering wheel heater switch

3. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step

4. If the cause is not visually evident, verify the symptom and refer to the Symptom Chart, alternatively check for Diagnostic Trouble Codes (DTCs) and refer to the DTC Index

5. Check DDW for open campaigns. Refer to the corresponding bulletins and SSMs which may be valid for the specific customer complaint and carry out the recommendations as required

SYMPTOM CHART

SYMPTOM	POSSIBLE CAUSES	ACTION
Steering wheel heater inhibited	- Incorrect power mode - Engine not running - Battery/charging system fault	**NOTE:** The steering wheel heater switch may illuminate amber even when the system is inhibited. - Start the engine and retest - Refer to the relevant section of the workshop manual and test the battery and charging system
Steering wheel heater inoperative	- LIN 1 circuit short circuit to ground, short circuit to power, open circuit, high resistance - Steering wheel heater control module power or ground circuit short circuit to	- GO to Pinpoint Test A.

	ground, short circuit to power, open circuit, high resistance ▪ Steering wheel heater switch circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Steering wheel heater circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Steering wheel heater control module internal failure	

PINPOINT TESTS

NOTE:

The steering wheel heater will not operate if the engine is not running.

NOTES:

- An infra-red thermometer is used during this test.
- The steering wheel temperature should have increased by approximately 10°C after 5 minutes.
- Do not measure the steering wheel temperature at the seam in the in the leather; this is where the heater element starts and finishes and will produce inaccurate test results.

NOTES:

- An infra-red thermometer is used during this test.
- The steering wheel temperature should stabilise at 30°C to 43°C.

PINPOINT TEST A : FUNCTION TESTS	
TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
A1: FUNCTION TEST 1	
	1 Start the engine
	2 Set the steering wheel heater switch to on
	Did the steering wheel heater switch illuminate amber? **Yes** **GO to A2**. **No** GO to Pinpoint Test **B.**
A2: FUNCTION TEST 2	
	1 Set the steering wheel heater switch to off
	2 Set the climate control to cool the vehicle interior for 5 minutes
	3 Using an infra-red thermometer, measure and record the temperature of the steering wheel rim at six equidistant points
	4 Set the steering wheel heater switch to on
	5 Using an infra-red thermometer, monitor the temperature of the steering wheel rim at six equidistant points
	Has the temperature increased at any point on the steering wheel rim? **Yes** **GO to A3**. **No** GO to Pinpoint Test **I.**
A3: FUNCTION TEST 3	
	1 Using an infra-red thermometer, monitor the temperature of the steering wheel rim at six equidistant points
	2 Wait for the temperature to stabilise
	Has the temperature stabilised within the specified range at all six points? **Yes** Steering wheel heater operating normally **No** GO to Pinpoint Test **J.**

PINPOINT TEST B : DIAGNOSTIC TROUBLE CODE TESTS	
TEST	DETAILS/RESULTS/ACTIONS

CONDITIONS	
B1: DIAGNOSTIC TROUBLE CODE TEST 1	
	1 Using the manufacturer approved diagnostic system, check the body control module for related DTCs (LIN or steering wheel heater DTCs)
	Are there any relevant DTCs set in the body control module? **Yes** GO to Pinpoint Test **C.** **No** GO to Pinpoint Test **H.**

WARNING:

Do not probe or disconnect the airbag connector(s) during this test.

WARNING:

Do not probe or disconnect the airbag connector(s) during this test.

WARNING:

Do not probe or disconnect the airbag connector(s) during this test.

PINPOINT TEST C : FASCIA CIRCUIT (MODULE) TESTS	
TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
C1: FASCIA CIRCUIT (MODULE) TEST 1	
	1 Remove the steering column lower cowl
	2 Set the ignition to on
	3 Set a multimeter to measure voltage
	4 Using the multimeter, measure and record the battery voltage (reference voltage)
	5 Connect the multimeter to clockspring connector C2LS41H terminals 7 and 8

	Does the multimeter display < battery voltage? **Yes** GO to C2. **No** GO to Pinpoint Test D.

C2: FASCIA CIRCUIT (MODULE) TEST 2

	1 Connect the multimeter to clockspring connector C2LS41H terminal 8 and earth
	Does the multimeter display > 0V? **Yes** Refer to the electrical circuit diagrams and check the steering wheel heater control module circuit (clockspring to earth) for open circuit, high resistance. Repair the harness as necessary **No** GO to C3.

C3: FASCIA CIRCUIT (MODULE) TEST 3

	1 Connect the multimeter across passenger junction box fuse 44
	Does the multimeter display > 0V? **Yes** GO to C5. **No** GO to C4.

C4: FASCIA CIRCUIT (MODULE) TEST 4

	1 Connect the multimeter to the battery positive terminal and passenger junction box fuse 44
	Does the multimeter display > 0V? **Yes** Refer to the electrical circuit diagrams and check the steering wheel heater control module circuit (battery positive terminal to fuse) for open circuit, high resistance. Repair the harness as necessary **No** Refer to the electrical circuit diagrams and check the steering wheel heater control module circuit (fuse to clockspring) for open circuit, high resistance. Repair the harness as necessary

C5: FASCIA CIRCUIT (MODULE) TEST 5

	1 Connect the multimeter across passenger junction box fuse 44
	2 Disconnect clockspring connector C2LS41H
	Does the multimeter display > 0V? **Yes** Refer to the electrical circuit diagrams and check the steering wheel heater control module circuit (fuse to clockspring) for short circuit to

	ground. Repair the harness as necessary **No** GO to Pinpoint Test **E.**

WARNING:

> Do not probe or disconnect the airbag connector(s) during this test.

PINPOINT TEST D : FASCIA CIRCUIT (LIN) TESTS	
TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
D1: FASCIA CIRCUIT (LIN) TEST 1	
	1 Connect the multimeter to clockspring connector C2LS41H terminal 4 and earth
	Does the multimeter display > 0V and < battery voltage? **Yes** GO to Pinpoint Test **G.** **No** **GO to D2**.
D2: FASCIA CIRCUIT (LIN) TEST 2	
	1 Disconnect clockspring connector C2LS41H
	2 Connect the multimeter to clockspring connector C2LS41H terminal 4 (harness side) and earth
	Does the multimeter display > 0V and < battery voltage? **Yes** GO to Pinpoint Test **F.** **No** Refer to the electrical circuit diagrams and check the LIN 1 circuit (body control module to clockspring) for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the harness as necessary

WARNING:

> Do not probe or disconnect the airbag connector(s) during this test.

PINPOINT TEST E : STEERING WHEEL (MODULE) SHORT CIRCUIT TESTS

TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
E1: STEERING WHEEL (MODULE) SHORT CIRCUIT TEST 1	
	1 Refer to the relevant section of the workshop manual and remove the driver airbag
	2 Check the integrity of the wiring harness (clockspring to steering wheel heater control module)
	Is a wiring harness fault present? **Yes** Repair the harness as necessary **No** GO to E2.
E2: STEERING WHEEL (MODULE) SHORT CIRCUIT TEST 2	
	1 Disconnect clockspring connector C2R115B
	2 Connect clockspring connector C2LS41H
	3 Refer to the relevant section of the workshop manual and install the driver airbag
	4 Connect the multimeter across passenger junction box fuse 44
	Does the multimeter display > 0V? **Yes** Install a new clockspring **No** Install a new steering wheel heater control module

WARNING:

Do not probe or disconnect the airbag connector(s) during this test.

PINPOINT TEST F : STEERING WHEEL (LIN) SHORT CIRCUIT TESTS	
TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
F1: STEERING WHEEL (LIN) SHORT CIRCUIT TEST 1	
	1 Refer to the relevant section of the workshop manual and remove the driver airbag
	2 Check the integrity of the wiring harness (clockspring to steering wheel heater control module)

	Is a wiring harness fault present? **Yes** Repair the harness as necessary **No** GO to F2.

F2: STEERING WHEEL (LIN) SHORT CIRCUIT TEST 2	
	1 Disconnect clockspring connector C2R115B
	2 Connect clockspring connector C2LS41H
	3 Refer to the relevant section of the workshop manual and install the driver airbag
	4 Connect the multimeter to clockspring connector C2LS41H terminal 4 and earth
	Does the multimeter display > 0V and < battery voltage? **Yes** Install a new steering wheel heater control module **No** Install a new clockspring

WARNING:

Do not probe or disconnect the airbag connector(s) during this test.

NOTE:

Rotate the steering wheel during this test to check for an intermittent fault.

WARNING:

Do not probe or disconnect the airbag connector(s) during this test.

NOTE:

Rotate the steering wheel during this test to check for an intermittent fault.

WARNING:

Do not probe or disconnect the airbag connector(s) during this test.

NOTE:

Rotate the steering wheel during this test to check for an intermittent fault.

PINPOINT TEST G : STEERING WHEEL (MODULE AND LIN) OPEN CIRCUIT TESTS	
TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
G1: STEERING WHEEL (MODULE AND LIN) OPEN CIRCUIT TEST 1	
	1 Refer to the relevant section of the workshop manual and remove the driver airbag
	2 Check the integrity of the wiring harness (clockspring to steering wheel heater control module)
	Is a wiring harness fault present? **Yes** Repair the harness as necessary **No** GO to G2.
G2: STEERING WHEEL (MODULE AND LIN) OPEN CIRCUIT TEST 2	
	1 Disconnect clockspring connector C2R115B
	2 Disconnect clockspring connector C2LS41H
	3 Set a multimeter to measure resistance
	4 Connect the multimeter to clockspring connector C2R115B terminal 4 and clockspring connector C2LS41H terminal 7

	Does the multimeter display > 0 ohms? **Yes** Install a new clockspring **No** GO to G3.
G3: STEERING WHEEL (MODULE AND LIN) OPEN CIRCUIT TEST 3	
	1 Connect the multimeter to clockspring connector C2R115B terminal 1 and clockspring connector C2LS41H terminal 8
	Does the multimeter display > 0 ohms? **Yes** Install a new clockspring **No** GO to G4.
G4: STEERING WHEEL (MODULE AND LIN) OPEN CIRCUIT TEST 4	
	1 Connect the multimeter to clockspring connector C2R115B terminal 2 and clockspring connector C2LS41H terminal 4
	Does the multimeter display > 0 ohms? **Yes** Install a new clockspring **No** Install a new steering wheel heater control module

WARNING:

Do not probe or disconnect the airbag connector(s) during this test.

PINPOINT TEST H : STEERING WHEEL HEATER SWITCH TESTS	
TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
H1: STEERING WHEEL HEATER SWITCH TEST 1	
	1 Refer to the relevant section of the workshop manual and remove the driver airbag
	2 Disconnect steering wheel heater switch connector C2LS41B
	3 Set a multimeter to measure resistance
	4 Connect the multimeter to steering wheel heater switch connector (switch side) C2LS41B terminals 3 and 9

	5 Operate the steering wheel heater switch
	Does the multimeter display change when the steering wheel heater switch is operated? **Yes** Install a new steering wheel heater control module **No** Install a new steering wheel heater switch

NOTE:

The steering wheel heater will not operate if the engine is not running.

NOTE:

The engine should still be running and the steering wheel heater set to on.

NOTE:

The engine should still be running and the steering wheel heater set to on.

NOTE:

The engine should still be running and the steering wheel heater set to on.

NOTE:

The engine should still be running and the steering wheel heater set to on.

WARNING:

Do not probe or disconnect the airbag connector(s) during this test.

NOTE:

The engine should still be running and the steering wheel heater set to on.

WARNING:

Do not probe or disconnect the airbag connector(s) during this test.

WARNING:

Do not probe or disconnect the airbag connector(s) during this test.

NOTE:

Rotate the steering wheel during this test to check for an intermittent fault.

WARNING:

Do not probe or disconnect the airbag connector(s) during this test.

NOTE:

Rotate the steering wheel during this test to check for an intermittent fault.

PINPOINT TEST I : STEERING WHEEL HEATER CIRCUIT TESTS	
TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
I1: STEERING WHEEL HEATER CIRCUIT TEST 1	
	1 Remove the steering column lower cowl
	2 Start the engine
	3 Set the steering wheel heater switch to on
	4 Set a multimeter to measure voltage
	5 Using the multimeter, measure and record the battery voltage (reference voltage)
	6 Connect the multimeter to clockspring connector C2R115D terminals 2 and 1
	Does the multimeter display < battery voltage? **Yes** **GO to I2**. **No** **GO to I10**.
I2: STEERING WHEEL HEATER CIRCUIT TEST 2	
	1 Connect the multimeter to clockspring connector C2R115D terminal 1 and earth
	Does the multimeter display > 0V? **Yes** Refer to the electrical circuit diagrams and check the steering wheel heater circuit (clockspring to earth) for open circuit, high resistance. Repair the harness as necessary **No**

	GO to I3.

I3: STEERING WHEEL HEATER CIRCUIT TEST 3

	1 Connect the multimeter across passenger junction box fuse 43P
	Does the multimeter display > 0V? **Yes** GO to I5. **No** GO to I4.

I4: STEERING WHEEL HEATER CIRCUIT TEST 4

	1 Connect the multimeter to the battery positive terminal and clockspring connector C2R115D terminal 2
	Does the multimeter display > 0V? **Yes** Refer to the electrical circuit diagrams and check the steering wheel heater circuit (battery positive terminal to fuse) for open circuit, high resistance. Repair the harness as necessary **No** Refer to the electrical circuit diagrams and check the steering wheel heater circuit (fuse to clockspring) for open circuit, high resistance. Repair the harness as necessary

I5: STEERING WHEEL HEATER CIRCUIT TEST 5

	1 Connect the multimeter across passenger junction box fuse 43P
	2 Set the steering wheel heater switch to off
	Does the multimeter display > 0V? **Yes** GO to I6. **No** GO to I9.

I6: STEERING WHEEL HEATER CIRCUIT TEST 6

	1 Connect the multimeter across passenger junction box fuse 43P
	2 Disconnect clockspring connector C2R115D
	Does the multimeter display > 0V? **Yes** Refer to the electrical circuit diagrams and check the steering wheel heater circuit (fuse to clockspring) for short circuit to ground. Repair the harness as necessary **No** GO to I7.

I7: STEERING WHEEL HEATER CIRCUIT TEST 7

	1 Refer to the relevant section of the workshop manual and remove the

	driver airbag
	2 Check the integrity of the wiring harness (clockspring to steering wheel heater control module)
	Is a wiring harness fault present? **Yes** Repair the harness as necessary **No** GO to I8.

I8: STEERING WHEEL HEATER CIRCUIT TEST 8

	1 Disconnect clockspring connector C2R115E
	2 Connect clockspring connector C2R115D
	3 Refer to the relevant section of the workshop manual and install the driver airbag
	4 Start the engine
	5 Connect the multimeter across passenger junction box fuse 43P
	Does the multimeter display > 0V? **Yes** Install a new clockspring **No** Install a new steering wheel heater control module

I9: STEERING WHEEL HEATER CIRCUIT TEST 9

	1 Refer to the relevant section of the workshop manual and remove the driver airbag
	2 Check the integrity of the wiring harness (clockspring to steering wheel heater control module)
	Is a wiring harness fault present? **Yes** Repair the harness as necessary **No** Install a new steering wheel

I10: STEERING WHEEL HEATER CIRCUIT TEST 10

	1 Refer to the relevant section of the workshop manual and remove the driver airbag
	2 Check the integrity of the wiring harness (clockspring to steering wheel heater control module)
	Is a wiring harness fault present? **Yes** Repair the harness as necessary

	No GO to I11.
	I11: STEERING WHEEL HEATER CIRCUIT TEST 11
	1 Disconnect steering wheel heater connector C2LS41K
	2 Set a multimeter to measure resistance
	3 Connect the multimeter to steering wheel heater connector C2LS41K (steering wheel side) terminals 1 and 4
	Does the multimeter display 2.25 ohms ± 10%? **Yes** GO to I12. **No** Install a new steering wheel
	I12: STEERING WHEEL HEATER CIRCUIT TEST 12
	1 Disconnect clockspring connector C2R115D
	2 Disconnect clockspring connector C2R115E
	3 Connect the multimeter to clockspring connector C2R115D terminal 2 and clockspring connector C2R115E terminal 2
	Does the multimeter display > 0 ohms? **Yes** Install a new clockspring **No** GO to I13.
	I13: STEERING WHEEL HEATER CIRCUIT TEST 13
	1 Connect the multimeter to clockspring connector C2R115D terminal 1 and clockspring connector C2R115E terminal 1
	Does the multimeter display > 0 ohms? **Yes** Install a new clockspring **No** Install a new steering wheel heater control module

PINPOINT TEST J : STEERING WHEEL PARTIAL FAILURE TESTS	
TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
	J1: STEERING WHEEL PARTIAL FAILURE TEST1
	1 Refer to the relevant section of the workshop manual and remove the driver airbag

	2 Disconnect steering wheel heater connector C2LS41K
	3 Set a multimeter to measure resistance
	4 Connect the multimeter to steering wheel heater connector C2LS41K (steering wheel side) terminals 1 and 4
	Does the multimeter display 2.25 ohms ± 10%? **Yes** Install a new steering wheel heater control module **No** Install a new steering wheel

DTC INDEX

For a list of Diagnostic Trouble Codes (DTCs) that could be logged on this vehicle, please refer to Section 100-00.

REFER to: Diagnostic Trouble Code Index - DTC: Body Control Module (BCM) (100-00 General Information, Description and Operation).